Carry out the following task as a part of the assignment. The dataset to use is The Breakfast Cereal dataset:

1. Bar Chart of Cereal Ratings:
Create a bar chart to visualize the cereal ratings, with each cereal represented by a bar. This will allow for easy comparison and identification of the highest and lowest rated cereals.

2. Scatter Plot of Sugar Content vs. Calorie Count:
Generate a scatter plot with sugar content on the x-axis and calorie count on the y-axis. Each data point represents a cereal, allowing for examination of the relationship between sugar content and calorie count.

3. Box Plot of Fiber Content by Cereal Type:
Create a box plot to visualize the distribution of fiber content for different cereal types (e.g., hot, cold). This will provide insights into the variability and central tendency of fiber content within each cereal type.

4. Bar Chart of Cereal Types:
Generate a bar chart to visualize the number of cereals in each cereal type category (e.g., hot, cold). This will provide an overview of the distribution of cereals across different types.

5. Heatmap of Nutritional Attributes:
Create a heatmap to display the nutritional attributes (e.g., sugar, protein, fiber) across different cereals. This will allow for easy identification of cereals with high or low values for each attribute.

6. Stacked Bar Chart of Sugar Content by Cereal Type:
Design a stacked bar chart to compare the sugar content of cereals within each cereal type. This will enable visualization of the sugar content distribution within different types of cereals.

7. Line Plot of Fiber Content Over Time:
If the dataset includes a time component (e.g., multiple years), create a line plot to visualize the change in fiber content over time for specific cereal brands or types.

8. Histogram of Calorie Count:
Generate a histogram to display the distribution of calorie counts across the cereals. This will provide insights into the spread and concentration of calorie values.

9. Grouped Bar Chart of Protein Content by Brand:
Create a grouped bar chart to compare the protein content of cereals across different brands. This will allow for a brand-wise comparison of protein content.

10. Pie Chart of Low-Fat vs. Non-Low-Fat Cereals:
Generate a pie chart to visualize the proportion of cereals classified as "low fat" versus those that are not. This will provide an overview of the distribution of low-fat cereals in the dataset.

Remember to label your axes, provide a title for each visualization, and use appropriate color schemes to enhance clarity and understanding. These visualizations will help uncover patterns, relationships, and insights within The Breakfast Cereal dataset, facilitating a deeper understanding of the nutritional attributes and characteristics of different cereals.